

PROJECT SYNOPSIS

AgroChain: Decentralized Verification and Peer-to-Peer Micro-Funding Infrastructure for Agricultural Cooperatives.

1. INTRODUCTION

Agriculture is one of the largest contributors to the Indian economy, yet farmers continue to face significant challenges regarding product authenticity, market transparency, and access to financial assistance. Existing agricultural supply chains rely heavily on centralized databases, making records vulnerable to tampering while reducing consumer trust. At the same time, small farmers struggle to obtain institutional loans due to collateral requirements and complicated banking procedures. Blockchain technology provides an immutable and transparent method of recording transactions. However, conventional blockchain applications require users to manage cryptocurrency wallets and transaction fees, making them difficult for farmers and ordinary consumers to adopt. AgroChain proposes a hybrid Web2/Web3 platform that combines the simplicity of traditional web applications with the security and immutability of blockchain. Farmers, consumers, and investors interact using a normal web application, while authorized Agricultural Inspectors and Quality Lab Testers utilize blockchain to record verification and certification details. The system also introduces a peer-to-peer microfinance marketplace where investors can directly support verified agricultural projects through Letter of Intent (LOI) proposals.

2. PROBLEM STATEMENT

Current agricultural supply chain systems suffer from several major limitations:

- Lack of transparency in crop tracking.
- Centralized databases can be modified without leaving audit trails.
- Consumers cannot verify the authenticity of agricultural products.
- Farmers have limited access to affordable financing.
- Verification records are scattered across different systems.
- Manual inspector assignment causes delays and inefficiencies.

These issues reduce trust among stakeholders and limit the growth of transparent agricultural commerce.

3. OBJECTIVES

The primary objectives of AgroChain are:

- To develop a blockchain-enabled agricultural traceability platform.
- To provide transparent crop lifecycle tracking from cultivation to sale.
- To create immutable quality certification using Ethereum blockchain.
- To develop a secure peer-to-peer agricultural investment platform.
- To automate inspector and laboratory assignment based on geographical location.

- To enable consumers to verify product authenticity through QR code scanning.
 - To provide a secure and user-friendly system without requiring cryptocurrency knowledge from farmers or consumers.
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4. EXISTING SYSTEM

Existing agricultural traceability systems mainly depend on centralized databases or private blockchain platforms.

Current solutions include:

- Enterprise blockchain solutions
- Cloud-based agricultural management systems
- Traditional agricultural lending platforms

Although these systems provide basic tracking and financing, they often require expensive infrastructure, lack public transparency, and depend on centralized authorities.

5. LIMITATIONS OF EXISTING SYSTEM

The major limitations are:

- Centralized data can be modified.
 - High implementation cost.
 - Limited consumer visibility.
 - Complex blockchain usability.
 - Lack of automated verification workflow.
 - Poor integration between financing and traceability.
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6. PROPOSED SYSTEM

AgroChain introduces a hybrid Web2/Web3 architecture where blockchain is used only for critical verification processes while routine operations are handled through a relational database for better performance.

The proposed system provides:

- Farmer registration
- Crop registration
- Agricultural inspection
- Laboratory quality certification
- Blockchain verification
- QR-based product traceability
- Peer-to-peer investment marketplace
- Administrative monitoring dashboard

Only authorized inspectors and quality testers interact with the blockchain using MetaMask, while other users access the platform through a normal web interface.

7. SCOPE OF THE PROJECT

Area	Description
Agricultural Crop Traceability	Tracks the complete lifecycle of crops from cultivation to consumer purchase.
Blockchain-Based Verification	Stores inspection and quality certification records on the Ethereum blockchain.
Farmer Registration	Enables farmers to register, manage profiles, and submit crop details.
Inspector Management	Assigns agricultural inspectors based on geographical location for crop verification.
Laboratory Certification	Allows quality laboratories to test and certify harvested crops.
Consumer Product Verification	Enables consumers to verify product authenticity using QR code scanning.
Peer-to-Peer Agricultural Funding	Provides a platform where investors can fund verified crops through Letter of Intent (LOI).
Administrative Monitoring	Allows administrators to manage users, monitor activities, and maintain system records.
Out of Scope	Logistics management, transportation tracking, and payment gateway integration are not included in the current implementation.

8. SYSTEM ARCHITECTURE

• Layer	• Components
• Client Layer	• React.js Frontend, Farmer Dashboard, Consumer Portal, Investor Portal, Inspector Dashboard, Admin Dashboard
• Application Layer	• Flask REST API, JWT Authentication, OTP Verification, Business Logic
• Database Layer	• PostgreSQL / SQLite, User Management, Crop Records, Investment Records, Audit Logs
• Blockchain Layer	• Ethereum, Solidity Smart Contracts, MetaMask Authentication, Immutable Verification Records

- Immutable Verification Records
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9. MODULES

• Module	• Functions
• User Authentication Module	• Registration, Login, OTP Verification, Role Management
• Farmer Module	• Crop Registration, Crop Status Updates, Document Upload
• Agricultural Inspector Module	• Crop Inspection, Blockchain Verification, Inspection Reports
• Quality Laboratory Module	• Product Testing, Quality Certification, Blockchain Approval
• Investor Module	• Browse Crops, Submit Letter of Intent (LOI), Investment Management
• Consumer Module	• QR Code Scanning, Crop Traceability, Product Verification
• Administrator Module	• User Management, Inspector Assignment, Analytics, Audit Monitoring

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10. METHODOLOGY

Step	Description
1	Farmer registers and uploads crop details.
2	System assigns a nearby Agricultural Inspector.
3	Inspector performs field verification.
4	Verification details are securely stored on the blockchain.
5	Farmer updates the crop harvest status.
6	Quality Laboratory performs product testing.
7	Laboratory issues a blockchain-based quality certificate.
8	Investor submits a funding proposal (LOI).
9	Farmer reviews and accepts the investment proposal.
10	Consumer scans the QR code to verify the complete product history.

11. TECHNOLOGY STACK

Layer	Technology
Frontend	React.js, Tailwind CSS
Backend	Python Flask
Database	PostgreSQL, SQLite
Blockchain	Ethereum
Smart Contracts	Solidity
Development	Hardhat
Authentication	JWT, OTP
Wallet	MetaMask
QR Scanner	html5-qrcode
Version Control	Git & GitHub

12. HARDWARE REQUIREMENTS

- Intel Core i3 or above
- 4 GB RAM minimum
- 500 MB Storage
- Internet Connection
- Webcam (optional for QR scanning)

13. SOFTWARE REQUIREMENTS

- Windows 10/11 or Linux
- Python 3.9+
- Node.js 18+
- Visual Studio Code
- PostgreSQL
- MetaMask
- Google Chrome
- Git

14. EXPECTED OUTCOME

The AgroChain platform is expected to:

- Improve transparency in agricultural supply chains.
- Prevent unauthorized modification of crop records.
- Increase consumer trust through QR-based verification.

- Simplify blockchain usage for non-technical users.
 - Provide secure digital quality certificates.
 - Enable peer-to-peer agricultural funding.
 - Improve efficiency of inspection and certification processes.
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15. FUTURE ENHANCEMENTS

Future enhancements include:

- AI-based crop disease detection.
 - IoT sensor integration for environmental monitoring.
 - Government land record integration.
 - Mobile application for field inspectors.
 - Digital payment gateway integration.
 - Real-time logistics and transportation tracking.
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16. CONCLUSION

AgroChain presents a hybrid Web2/Web3 solution for improving transparency, trust, and financial accessibility in agricultural supply chains. By combining blockchain technology with a conventional web application architecture, the system ensures secure verification without increasing complexity for end users. The proposed platform strengthens consumer confidence, improves supply chain integrity, and creates new funding opportunities for farmers through decentralized yet user-friendly technology.
